

HDI Predictor Documentation

Technology Stack

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Flask Jinja2 templates)	Lightweight, form-driven UI is enough for a data-entry-and-result tool; no need for a heavy SPA framework
2	Backend / Server-Side	Python 3, Flask	Fast to build a small prediction API; Python has first-class ML tooling (scikit-learn) so the same language handles both serving and modeling
3	Machine Learning	Scikit-learn (RandomForestClassifier), Pandas, NumPy	RandomForest handles the mixed-scale numeric indicators well, resists overfitting on a modest dataset, and gives built-in feature importances for explainability
4	Data Storage	CSV dataset (country-level HDI indicators) + Pickled model artifact (model.pkl)	The dataset is small and tabular, so a flat file is sufficient; the trained model is serialized once and reloaded at runtime for fast inference
5	Cloud / Hosting / Deployment	Docker	Containerization guarantees the app runs identically regardless of the evaluator's local Python/OS setup, avoiding "works on my machine" issues
6	Version Control & CI/CD	Git & GitHub	Enables versioned, collaborative development and straightforward

			sharing of the repository for evaluation
7	Development Environment	Jupyter Notebook (model experimentation), VS Code (application code)	Notebooks are ideal for iterative EDA/model tuning; VS Code is used for writing and structuring the production Flask app